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Adjustable leveling kit and associated installation method

Abstract

A kit includes an adjustable chock provided with a first component having screw threads and with a second component having screw threads cooperating with the screw threads of the first component and having a lower bearing surface, and with a bearing element provided with a lower bearing surface in contact with an upper bearing surface of the first component, and with an upper bearing surface. The kit further includes at least one friction ring adapted to be mounted against the lower bearing surface of the second component or against the upper bearing surface of the bearing element.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application claims priority to German Patent Application no. 102021207060.4, filed Jul. 6, 2021, the contents of which is fully incorporated herein by reference.
TECHNICAL FIELD OF THE INVENTION
(2) The present invention relates to a system configured for use as an adjustable kit for levelling and anchoring a frame of a machine to a support.
BACKGROUND OF THE INVENTION
(3) Adjustable levelling pads or chocks are generally configured to provide both support and vertical alignment capability with or without an associated anchor bolt.
(4) Adjustable chocks are well known in the art.
(5) Reference can be made to FIGS. **1**A and **1**B which illustrates a known adjustable chock **10**.
(6) The adjustable chock **10** is mounted to connect the frame **1** of a machine to a foundation or support **2**, for example constructed from concrete or steel. Anchoring the frame **1** of the machine to the support **2** is here done with an anchor bolt **3**.
(7) The adjustable chock **10** comprises a first component **11** or shaft element, a second component **12** or annular element and a third component **13** or bearing element. The first, second and third components **11**, **12**, **13** are coaxial along a vertical axis Z-Z′.
(8) The first component **11** comprises an upper portion **11***a* and a lower portion **11***b* provided with an outer screw thread **11***c*. As illustrated on FIG. **1**B, the upper portion **11***a* has a partly upper surface **11***d* of concave shape. The first component **11** has a through-hole **14** for accommodating a shank **3***a* of the bolt **3**.
(9) The second component **12** has a second through-hole **12***a* provided with an inner screw thread **12***b* configured to engage with the outer screw thread **11***c* of the first component **11**. The threaded portions **11***c*, **12***b* cooperate together and provide a vertical adjustment.
(10) The third component **13** sits between the frame **1** of the machine and the upper portion **11***a* of the first component **11**.
(11) As shown in FIG. **1**B, the third component **13** has a lower surface **13***a* engaging with the upper surface **11***d* of the first component **11**. The lower surface **13***a* and the upper surface **11***d* are

complementarily shaped so as to facilitate slight adjustment of the positions between the first component **11** and the third component **13** relative to one another, for example, in order to accommodate slight deviations from the frame **1** of the machine and the support **2**.
(12) The third component **13** has a through hole **16** having a diameter larger than the diameter of the first through-hole **14** in order to allow the shank **3***a* of the bolt **3** to pass through if an axis of symmetric of the lower surface **11***d* of the first component **11** is not aligned with an axis of symmetry of the lower surface **13***a* of the third component **13**. This leads to accommodate deviations from horizontal, parallel orientations of the frame **1** of the machine and the support **2**.
(13) The chock **10** is sandwiched between the frame **1** of the machine and the support **2** and securely held in place by the bolt **3** and a nut **4** screwed on a part of the shank **3***a* extending beyond the frame **1** of the machine. The height of the adjustable chock **10** is adjusted by screwing the first component **11** further into or further out of the second component **12**.
(14) When installed, the chock **10** is subjected to a mechanical load as a result of the weight of the frame **1** of the machine, and also as a result of reaction forces transmitted by the support **2** and/or by the frame **1**.
(15) The axial stiffness of an adjustable chock **10** depends from the axial load capacity and the transverse stiffness depends from transverse load capacity. The axial stiffness can be influenced by the dimensions of the chock **10**, mainly by the thread connection between the first and second components **11**, **12**.
(16) The aim of the present invention is to increase the transverse load capacity of an adjustable chock.
SUMMARY OF THE INVENTION
(17) The invention relates to a kit comprising an adjustable chock provided with a first component having screw threads, with a second component having screw threads cooperating with the screw threads of the first component and having a lower bearing surface, and with a bearing element provided with a lower bearing surface in contact with an upper bearing surface of the first component, and with an upper bearing surface.
(18) The kit further comprises at least one friction ring adapted to be mounted against the lower bearing surface of the second component of the chock, or against the upper bearing surface of the bearing element of the chock.
(19) Accordingly, the friction coefficient between the second component of the chock and the associated support, or the friction coefficient between the bearing element of the chock and the associated machine is increased. This leads to increase the transverse load capacity of the adjustable chock, i.e. the resistance to move in the horizontal direction.
(20) Preferably, the friction coefficient of the at least one friction ring is at least equal to zero point six five (0.65).
(21) Preferably, the kit comprises a first friction ring adapted to be mounted against the lower bearing surface of the second component of the chock, and a second friction ring adapted to be mounted against the upper bearing surface of the bearing element of the chock.
(22) The first friction ring may be sized to cover the entire lower bearing surface of the second component of the chock.
(23) The second friction ring may be sized to cover the entire upper bearing surface of the bearing element of the chock.
(24) Preferably, the lower and upper frontal faces of the first friction ring are provided with a coating having at least one layer and diamond particles embedded in the layer.
(25) Preferably, the lower and upper frontal faces of the second friction ring are provided with a coating having at least one layer and diamond particles embedded in the layer.
(26) For example, the mean diamond particles size may be comprised between 10 μm and 55 μm. The diamond particles concentration on the surface of the layer may be comprised between 8% and 60%.

(27) In one embodiment, the at least one friction ring is a friction shim, for example the friction shim commercialized by the company 3M or by the company Atela.
(28) Alternatively, the at least one friction ring may be a friction disc commercialized by the applicant.
(29) In one embodiment, the friction ring(s) and the chock of the kit are delivered as separate parts. Alternatively, the kit may be delivered as a pre-assembled unit. In this case, each friction ring is mounted against the bearing surface of the associated component of the chock and also secured onto this bearing surface.
(30) The invention also relates to an installation method of the kit as previously defined for levelling and anchoring a machine to a support, wherein the at least one friction ring is axially interposed between the second component and the support of the chock, or axially interposed between the bearing element of the chock and the machine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) At least one of the embodiments of the present invention is accurately represented by this application's drawings which are relied on to illustrate such embodiment(s) to scale and the drawings are relied on to illustrate the relative size, proportions, and positioning of the individual components of the present invention accurately relative to each other and relative to the overall embodiment(s). Those of ordinary skill in the art will appreciate from this disclosure that the present invention is not limited to the scaled drawings and that the illustrated proportions, scale, and relative positioning can be varied without departing from the scope of the present invention as set forth in the broadest descriptions set forth in any portion of the originally filed specification and/or drawings. The present invention and its advantages will be better understood by studying the detailed description of specific embodiments given by way of non-limiting examples and illustrated by the appended drawings on which:
(2) FIG. **1**A is a perspective view of a known adjustable chock,
(3) FIG. **1**B shows a partial cross-section of the adjustable chock of FIG. **1**A in operational use, and
(4) FIG. **2** is a cross-section view of a kit according to an example of the invention, in operational use.
DETAILED DESCRIPTION OF THE INVENTION
(5) Those of ordinary skill in the art will appreciate from this disclosure that when a range is provided such as (for example) an angle/distance/number/weight/volume/spacing being between one (1 of the appropriate unit) and ten (10 of the appropriate units) that specific support is provided by the specification to identify any number within the range as being disclosed for use with a preferred embodiment. For example, the recitation of a percentage of copper between one percent (1%) and twenty percent (20%) provides specific support for a preferred embodiment having two point three percent (2.3%) copper even if not separately listed herein and thus provides support for claiming a preferred embodiment having two point three percent (2.3%) copper. By way of an additional example, a recitation in the claims and/or in portions of an element moving along an arcuate path by 3 at least twenty (20°) degrees, provides specific literal support for any angle greater than twenty(20°) degrees, such as twenty-three(23°) degrees, thirty(30°) degrees, thirty-three-point five(33.5°) degrees, forty-five(45°) degrees, fifty-two(52°) degrees, or the like and thus provides support for claiming a preferred embodiment with the element moving along the arcuate path thirty three-point five(33.5°) degrees. As shown on FIG. **2**, the kit comprises an adjustable chock **100** mounted to connect a frame **1** of a machine to a foundation or support **2**. Anchoring the frame **1** of the machine to the support **2** is here done with an anchor bolt **3**.
(6) The adjustable chock **100** comprises a first component **110** or shaft element, a second

component **120** or lower adjustable part and a third component or bearing element **130**. The first, second and third components **110**, **120**, **130** are coaxial along a vertical axis Z-Z′. The chock **100** is symmetrical relative to the longitudinal axis Z-Z′. The first, second and third components **110**, **120**, **130** are made, for example, of steel, preferably high-grade steel.
(7) As will be described latter, the kit further comprises first and second friction shims **140**, **150** respectively axially mounted between the second component **120** of the chock and the support **2**, and between the bearing element **130** of the chock and the frame **1** of the machine. The first and second friction shims **140**, **150** and the chock **100** of the kit may be delivered as three separate parts.
(8) The first component **110** of the adjustable chock comprises a lower portion **111** and an upper flange **112**. The first component **110** is provided with an outer screw thread **111***b*. The outer screw thread **111***b* is formed on the outer surface of the lower portion **111**. The upper flange **112** protrudes radially outwards with respect to the lower portion **111**. The upper flange **112** has an upper bearing surface **113** at least partly of upwardly concave shape. The upper surface **113** is rotationally symmetrical. The upper surface **113** forms the upper surface of the first component **110**.
(9) The first component **110** has a first through-hole **115** extending axially from the upper surface **113** to a lower surface **114** of the first component **110**. The first through-hole **115** has an inner diameter configured for accommodating a shank **3***a* of the bolt **3**. The bolt **3** comprises the shank **3***a* and a threaded part **3***b*, for fitted bolts, the shank **3***a* having a diameter bigger than the diameter of the threaded part **3***b*.
(10) The second component **120** of the adjustable chock is provided with a lower bearing surface **121** and an upper surface **122**. The lower and upper surfaces **121**, **122** axially delimit the second component **120**. The lower surface **121** is axially opposite to the upper surface **122**. The upper surface **122** is located axially on the side of the upper flange **112** of the first component. The second component **120** is also provided with an inner bore and with an outer surface (not referenced) which are axially delimited by the lower and upper surfaces **121**, **122**. The bore is provided with inner screw threads **123** configured to engage with the outer screw threads **111***b* of the first component **110**. The screw threads **111***b*, **123** cooperate together and provide a vertical adjustment.
(11) The first component **110** is movable with respect to the second component **120** between a partially screwed position, shown on FIG. **2**, in which the threads **111***b* of the first component **110** partially cooperate with the threads **123** of the second component **120** and a totally screwed position, not shown, in which the lower surface of the upper flange **112** of the first component axially abuts against the upper surface **122** of the second component.
(12) As previously mentioned, the first friction shim **140** of the kit is axially mounted between the second component **120** of the adjustable chock and the support **2**. The friction shim **140** is axially mounted between the lower surface **121** of the second component and the support **2**. The friction shim **140** has an annular form. The friction shim **140** covers the entire lower surface **121** of the second component. The radial dimension of the friction shim **140** is substantially equal to the radial dimension of the lower surface **121**.
(13) The friction shim **140** is provided with opposite lower and upper frontal faces which axially delimit the axial thickness of the friction shim. The lower frontal face of the friction shim **140** is in axial contact with the support **2**. The upper frontal face of the friction shim **140** is in axial contact with the second component **120**.
(14) Preferably, the friction coefficient of the friction shim **140** is at least equal to zero point six five (0.65). For example, the friction shim **140** may comprise a steel substrate and two opposite coating each having a layer, for example a nickel layer, and diamond particles embedded in the layer. For example, the thickness of the steel substrate may be equal to zero point one (0.1) mm. The mean diamond particles size may be comprised between 10 μm and 55 μm. The diamond concentration on the surface layer may be comprised between 8% and 60%. The thickness of each

layer may be comprised between 5 μm and 30 μm. Each of the lower and upper frontal faces of the friction shim **140** is formed by the layer and the protruding diamond particles of the coating.
(15) The bearing element **130** of the adjustable chock sits on the first component **110**. The bearing element **130** sits on the upper flange **112** of the first component **110**. The bearing element **130** is provided with a lower bearing surface **131** and an upper bearing surface **132**. The lower surface **131** is axially opposite to the upper surface **132**. The lower surface **131** is in contact with the upper surface **113** of the first component. The lower surface **131** has a convex shape and is rotationally symmetrical.
(16) The lower surface **131** and the upper surface **113** are complementarily shaped so as to facilitate slight adjustment of the positions between the first component **110** and the bearing element **130** relative to one another, for example, in order to accommodate slight deviations from the frame **1** of the machine and the support **2**.
(17) The radius of curvature of the lower surface **131** of the bearing element **130** corresponds to the radius of curvature of the upper surface **113** of the first component **110**. In the illustrated example, the upper bearing surface **132** of the bearing element **130** extends radially.
(18) As previously mentioned, the second friction shim **150** of the kit is axially mounted between the bearing element **130** of the adjustable chock and the frame **1** of the machine. The friction shim **150** is axially mounted between the upper bearing surface **132** of the bearing element and the frame **1**. The friction shim **150** sits between the frame **1** and the bearing element **130**. The friction shim **150** has an annular form. The friction shim **150** covers the entire upper bearing surface **132** of the bearing element. The radial dimension of the friction shim **150** is substantially equal to the radial dimension of the upper bearing surface **132**.
(19) The friction shim **150** is provided with opposite lower and upper frontal faces which axially delimit the axial thickness of the friction shim. The lower frontal face of the friction shim **150** is in axial contact with the bearing element **130**. The upper frontal face of the friction shim **150** is in axial contact with the frame **1** of the machine.
(20) Preferably, the friction coefficient of the friction shim **150** is at least equal to zero point six five (0.65). Similarly to the friction shim **140**, the friction shim **150** may for example comprise a steel substrate and two opposite coating each having a layer, for example a nickel layer, and diamond particles embedded in the layer. For example, the thickness of the steel substrate may be equal to zero point one (0.1) mm. The mean diamond particles size may be comprised between 10 μm and 55 μm. The diamond concentration on the surface layer may be comprised between 8% and 60%. The thickness of each layer may be comprised between 5 μm and 30 μm. Each of the lower and upper frontal faces of friction shim **150** is formed by the layer and the protruding diamond particles of the coating.
(21) The bearing element **130** of the adjustable chock and the friction shim **150** are able to move with respect to the first component **110** allowing the inclination of the upper frontal face of the friction shim **150** to be adjusted with respect to the bottom surface of the frame **1** of the machine, so that flat contact of the lower frontal face of the friction shim **140** on the support **2** can be achieved, as well as flat contact of the upper frontal face of the friction shim **150** with the bottom surface of the frame **1** of the machine to be supported.
(22) As illustrated, the kit is sandwiched between the frame **1** of the machine and the support **2** and securely held in place by the bolt **3** and a nut **4** screwed on a part of the shank **3***a* extending beyond the machine **1**.
(23) The height H of the kit is adjusted between a minimal total height and a maximal total height by means of screwing the first component **110** further into or further out of the second component **120**. Indeed, by rotating the first component **110** with respect to the second component **120**, the vertical distance bridged by the kit can be set as desired.
(24) During the installation of the kit between the frame **1** of the machine and the support **2**, the friction ring **140** is axially interposed between the second component **120** of the adjustable chock

and the support **2**, and the friction ring **140** is axially interposed between the bearing element **130** of the adjustable chock and the frame **1**. There is no direct contact between the adjustable chock **100** and between the frame **1** and the support **2**.

(25) In the illustrated example, the first component **110** of the adjustable chock is provided with an outer screw thread **111***b* and the second component **120** is provided with an inner screw thread **123**. Alternatively, the second component **120** may be provided with an outer screw thread cooperating with an inner screw thread of the first component **110**.

Claims

1. A kit adapted for use in an adjustable chock, the kit comprising: a first component having a first component upper bearing surface, the first component upper bearing surface comprising a flat, axially outermost, radially extending portion; a second component threadably engaged with the first component, the second component having a second component lower bearing surface and a second component upper bearing surface, the second component upper bearing surface comprising a radially outermost portion and a radially innermost annular flat portion, the radially outermost portion of the second component upper bearing surface sloping such that an axial length of the second component decreases from a radially outermost point of the radially innermost annular flat portion of the second component upper bearing surface to a radially outermost point of the second component upper bearing surface; a bearing element provided with a bearing element lower bearing surface in contact with the first component upper bearing surface, the bearing element lower bearing surface not being in contact with the flat, axially outermost, radially extending portion, the bearing element also having a bearing element upper bearing surface; and a friction ring adapted to be mounted against either one of the groups of the second component lower bearing surface and the bearing element upper bearing surface in a way that eliminates a direct contact of the either one of the group of the second component lower bearing surface and the bearing element upper bearing surface with a corresponding one of the group of a supporting surface and a supported surface.
2. The kit according to claim 1, wherein a friction coefficient of the friction ring is at least zero point six five (0.65).
3. The kit according to claim 1, wherein the friction ring comprises a steel substrate.
4. A kit adapted for use in an adjustable chock, the kit comprising: a first component having a first component upper bearing surface; a second component threadably engaged with the first component and having a second component lower bearing surface; a bearing element provided with a bearing element lower surface in contact the first component upper bearing surface, the bearing element having a bearing element upper bearing surface; a first friction ring adapted to be mounted against the second component lower bearing surface; and a second friction ring adapted to be mounted against the bearing element upper bearing surface.
5. The kit according to claim 4, wherein the first friction ring is configured to cover the entire second component lower bearing surface.
6. The kit according to claim 4, wherein the second friction ring is configured to cover the entire bearing element upper bearing surface.
7. The kit according to claim 4, wherein the first friction ring has lower and upper first friction ring frontal faces, the lower and upper first friction ring frontal faces each provided with a coating, the coating comprising a layer having a plurality of diamond particles embedded therein.
8. The kit according to claim 7, wherein the plurality of diamond particles has a mean size of between ten micrometers (10 μm) and fifty-five micrometers (55 μm).
9. The kit according to claim 7, wherein the plurality of diamond particles occupy between eight percent (8%) and sixty percent (60%) of the layer.
10. The kit according to claim 4, wherein the second friction ring has lower and upper second

friction ring frontal faces, the lower and upper second friction ring frontal faces each provided with a coating comprising a layer having a plurality of diamond particles embedded therein.

11. The kit according to claim 4, wherein the first friction ring comprises a steel substrate.

12. The kit according to claim 11, wherein the first friction ring comprises two opposite coatings each having a nickel layer with a plurality of diamond particles embedded therein.

13. The kit according to claim 12, wherein the second friction ring comprises a steel substrate.

14. The kit according to claim 13, wherein the second friction ring comprise two opposite coatings each having a nickel layer with a plurality of diamond particles embedded therein.

15. A method for levelling and anchoring a machine to a support, the method comprising the steps of: providing a first component having a first component upper bearing surface; threadably engaging a second component with the first component, the second component having a second component lower bearing surface; positioning a bearing element in contact with the first component such that a bearing element lower surface contacts the first component upper bearing surface, the bearing element having a bearing element upper bearing surface; positioning first friction ring against the second component lower bearing surface; and positioning a second friction ring against the bearing element upper bearing surface.

16. The method according to claim 15, further comprising the step of joining the machine to the support via a fastener which extends through the first friction ring, the second component, the first component, the bearing element, and the second friction ring.

17. The method according to claim 16, adjusting a relative position of the first component and the second component.

18. The method according to claim 16, adjusting a relative position of the first component and the bearing element to level the machine.

19. The method according to claim 16, tightening the fastener so as to secure the relative position between the first component and the second component and to secure the relative position between the first component and the bearing element.

20. A kit adapted for use in an adjustable chock, the kit comprising: a first component having a first component upper bearing surface; a second component threadably engaged with the first component, the second component having a second component lower bearing surface; a bearing element provided with a bearing element lower bearing surface in contact with the first component upper bearing surface, the bearing element also having a bearing element upper bearing surface; and a friction ring adapted to be mounted against either one of the group of the second component lower bearing surface and the bearing element upper bearing surface, the friction ring being metallic, wherein the friction ring comprises a steel substrate, wherein the friction ring comprises two opposite coatings each having a nickel layer with a plurality of diamond particles embedded therein.